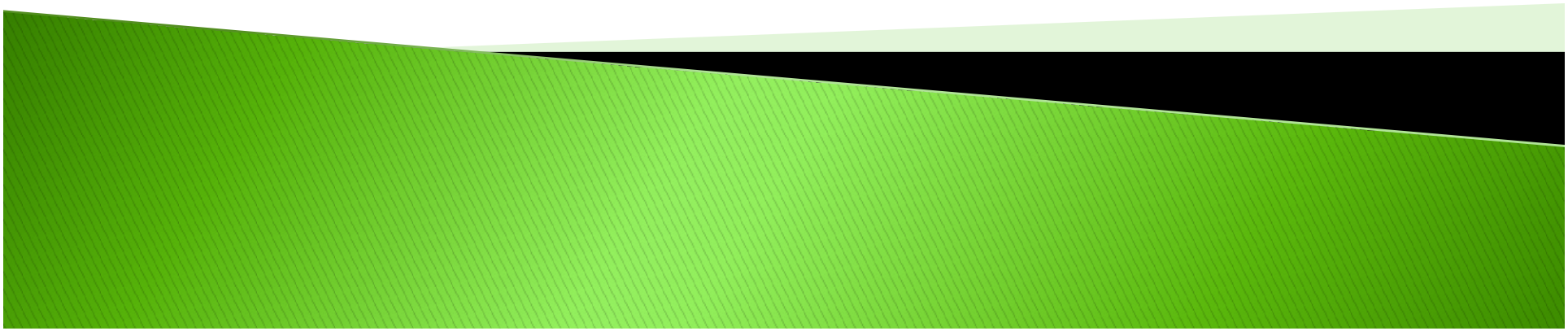
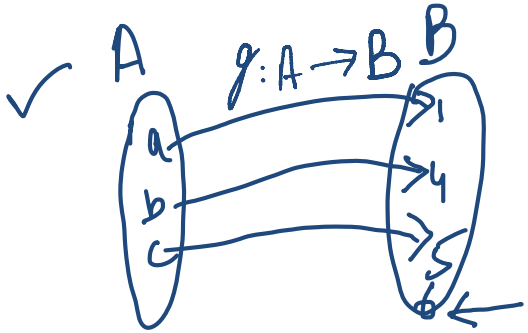
Ordering of Cardinal Numbers



Definition

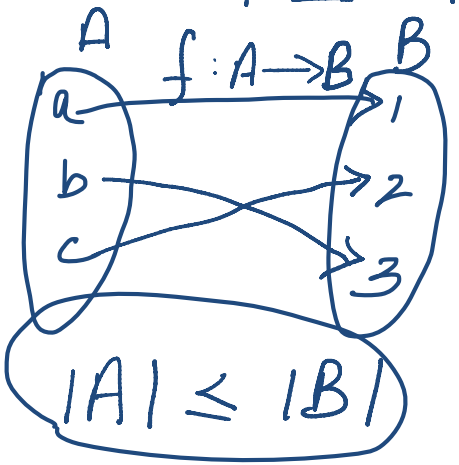
Let A and B be two sets. We say that $|A| \leq |B|$ if there exists a one-to-one(injective) function $f : A \to B$.

The above relation between cardinal numbers are well defined, that is, the relations are independent of the particular sets involved. Namely, if $A \approx A'$ and $B \approx B'$, $|A| \leq |B|$ iff $|A'| \leq |B'|$ $|A| = |B|$



$|A| < |B|$

Here g is one-to-one.
but g is not onto.



Example 1

Let ~~A~~ B be a proper subset of a finite set ~~B~~ A.

For example, $A = \{1,2,3,4\}$, then $B = \{1,2\}$

Here $|B| < |A|$.

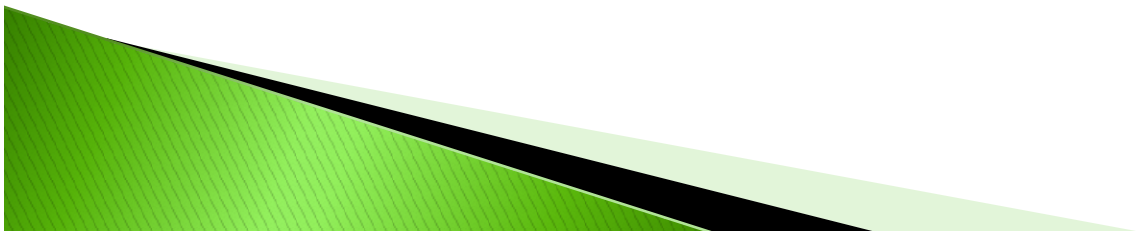
Example 2

Let n be a finite cardinal. Let A be a finite subset of $P = \{1,2,3,\cdots\}$. Then $|A| \neq |P|$. Thus $|A| < |P|$.

$n < \aleph_0$

We may write

$$0 < 1 < 2 < \cdots < \aleph_0$$



Example 3

Consider the set P of positive integers and the unit interval I, that is, consider the sets

$P = \{1, 2, 3, \ldots\}$ and $I = [0, 1] = \{x \in R : 0 < x \leq 1\}$

The function $f : P \rightarrow I$ defined by $f(n) = 1/n$ is one-to-one.(prove it.)

Let $f(n_1) = f(n_2)$

$\Rightarrow 1/n_1 = 1/n_2$

$\Rightarrow n_1 = n_2$ Proved

$f(1) = 1/1 = 1 \in I$

$f(2) = 1/2 = 0.5 \in I$

$f(3) = 1/3 = 0.3.. \in I$

$f(4) =$ different

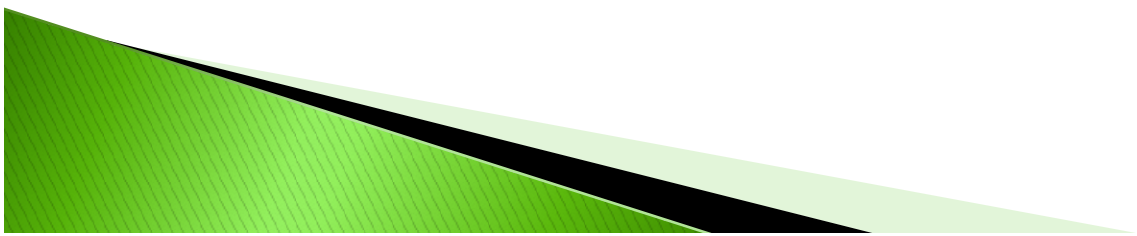
$|P| = \aleph_0$, $|I| = c$

Therefore, $|P| \leq |I|$. ✓

On the other hand $|P| \neq |I|$. Therefore $|P| < |I|$.

Accordingly we may write $\Rightarrow \aleph_0 < c$.

$0 < 1 < 2 < \cdots < \aleph_0 < c$.



Example 4

Let A be any infinite set. So A contains a subset which is denumerable. Accordingly, for any infinite set A, we always have

$\aleph_0 \leq |A|$.

$P = \{1, 2, 3, \ldots\}$

$E \subseteq P$.

$|P| = |E|$

$\aleph_0 = \aleph_0$

$\mathbb{R}$ = infinite set,

$P \subseteq \mathbb{R}$.

Here P is denumerable.

$|P| \leq |\mathbb{R}|$

$\aleph_0 < c$.